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Deformable image registration using multi-resolution vision Transformer for cardiac motion estimation

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E-mail: xslu-scuec@hotmail.com, suzhang@sjtu.edu.cn and xieqinlan@126.com**Keywords:** deformable registration, vision Transformer, multi-resolution optimization, cardiac magnetic resonance, motion estimation

Abstract

Objective. Deformable registration plays a crucial role in motion estimation from a sequence of cardiac magnetic resonance (CMR) imaging, which is good for the diagnosis and treatment of heart diseases. To address the challenges posed by intensity inhomogeneity and complex deformation, we propose a **novel convolutional neural network-Transformer framework** for this task. **Approach.** In this study, a convolutional projection Transformer block that enables efficient self-attention computation was designed for modeling long-range spatial correspondences. Additionally, a cooperative learning pattern was adopted for fusing information from global and local features. Finally, **multi-resolution strategy was employed for optimizing model parameters in coarse-to-fine manner.** **Main result.** The proposed method was evaluated on **three different CMR datasets for intra-subject registration.** Experimental results show that the proposed method **achieves better Dice overlap and lower surface distance**, compared to four non-learning-based methods and three deep-learning-based methods. **Significance.** For the challenging task of CMR image registration, our method demonstrates superior performance, delivering more accurate results with lower complexity. It may thus facilitate cardiac motion estimation for clinical assessments of cardiac function.

1. Introduction

Cardiac magnetic resonance (CMR) imaging is an important means to visualize the heart, which can provide diagnosis and prognosis information for patients with cardiovascular disease (Roos and Higgins 2014). Cardiac motion can be observed from the balanced steady-state free-precession (bSSFP) cine sequence of CMR, especially for the **contracting and relaxing pattern of the myocardium** under spatial and temporal resolution (Berger *et al* 2015). Depending on advanced technologies of medical image analysis, the **motion of the heart can be characterized by the regional deformation parameters**, such as **volume output and strain**. The analysis of such parameters has been shown to assist the diagnosis and treatment of heart diseases (Tobon-Gomez *et al* 2013).

At present, the status of cardiac motion is often assessed by visual inspection in clinical practice. It is time-consuming and subject to inter- and intra-observer (intra-obs) variations. Automatic estimation of cardiac motion is highly desired to generate relevant quantitative parameters. However, some factors make it **challenging** to obtain the motion estimation from the cine MR images over a cardiac cycle. First, **intensity inhomogeneities** can be found in the left ventricles of **images between different time phases**, because of the presence of papillary muscles and trabeculations. Also, **slice misalignments** during cine MR image acquisition lead to **large and complex deformations** within each subject (Dangi *et al* 2018).

In previous attempts, most studies in cardiac motion estimation have focused on three categories of methods (Wang and Amini 2012): feature tracking methods, optical flow methods, and registration-based methods. In general, the goal of feature tracking methods (Yan et al 2007, Obokata et al 2015, Zhang et al 2019) is to find the transformation by the correspondences of feature points on the boundaries. This type of method is often involved in segmenting the ventricular wall, such as epicardial and endocardial boundaries. The fundamental assumption of optical flow methods firstly proposed in computer vision (Horn and Schunck 1981) is that image intensity remains constant along a motion trajectory. Hassanein et al (2015) applied a band-pass optical flow technique to measure Eulerian strain from tagged CMR images. Sun et al (2018) presented a compact but effective convolutional neural network (CNN) model, called PWC-Net for optical flow. Nevertheless, it is difficult for the cine MR images that the intensity constancy condition is satisfied throughout the cardiac cycle.

The registration-based methods consider cardiac motion tracking as the problem of searching the optimal transformation between time phases from cine MR. Lamacie et al (2017) adopted the deformable registration algorithms for myocardial strain assessment in comparison with feature tracking and speckle-tracking echocardiography. Traditional registration optimizes an energy function in an iterative manner for each image pair, which need relatively long computation time. Recently, some researchers optimized the energy function by means of CNN, which can learn the common representation of image registration from training images. Vos et al (2019) proposed the deep learning image registration framework for unsupervised affine and deformable image registration, where the trained CNN can be used to register pairs of unseen images in one shot. For the task of cardiac motion estimation, the advantage of unsupervised CNN-based methods is taking orders-of-magnitude less time of registration that is available for real-time scenarios (Morales et al 2019). To further reduce local minima, a deep Laplacian pyramid network was presented throughout the coarse-to-fine optimization scheme for deformable image registration (Mok and Chung 2020). To overcome the limitation of CNN in modeling explicit long-range spatial relations, a hybrid Transformer-CNN model was developed for volumetric medical image registration (Chen et al 2022). Nevertheless, registration combining Transformer with multi-resolution optimization strategy remains to be investigated.

In this paper, we explored the CNN-Transformer hybrid framework of deformable image registration with the multi-resolution scheme for cardiac motion estimation. To be specific, a convolutional projection Transformer block (CPTB) substituting convolution operation for patch embedding and linear mapping was proposed for efficient self-attention computation. Then, the vision Transformer (ViT) was embedded into the Laplacian pyramid network in order to maintain the non-linearity of feature maps. We also introduced a global and local cooperative training pattern for deformable registration.

2. Methods

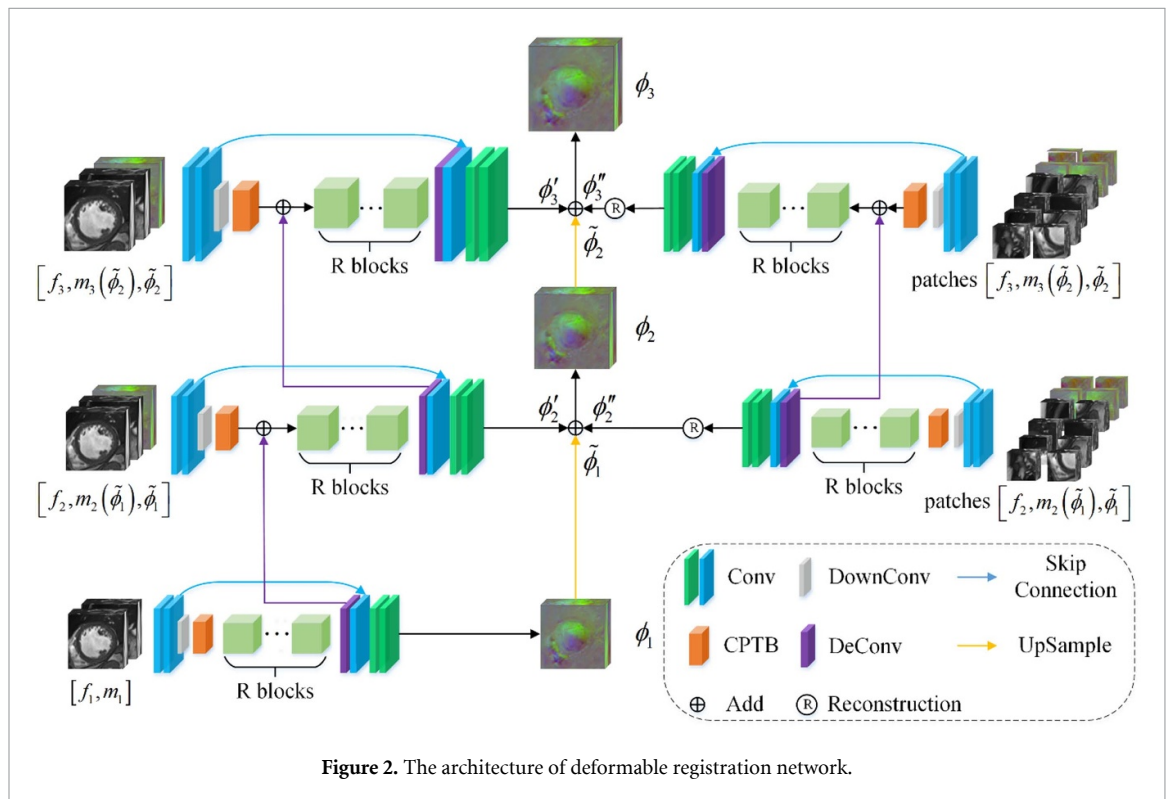
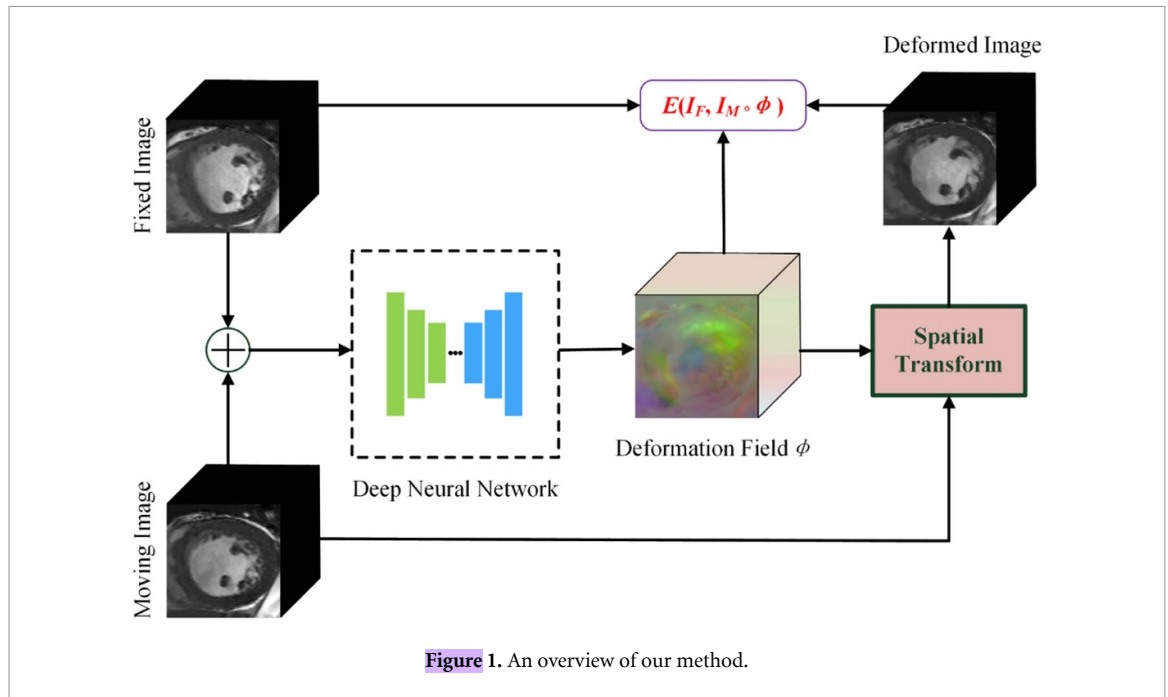
Let I_F and I_M be respectively the fixed image and moving image that are each defined on their own d -dimensional spatial domain $\Omega_F \subset \mathbb{R}^d$ and $\Omega_M \subset \mathbb{R}^d$. Deformable image registration is to establish spatial correspondence between I_F and I_M by optimizing an energy function E :

$$\hat{\phi} = \arg \min_{\phi} E(I_F, I_M \circ \phi), \quad (1)$$

where ϕ denotes the deformation field. In contrast to conventional registration, deep neural networks can be used to optimize the energy function, which maps an input image pair to a deformation field. Subsequently, a common registration representation can be learnt from a training dataset and then predict the deformation fields of unseen image pairs. Following this strategy, an overview of our method is shown in figure 1.

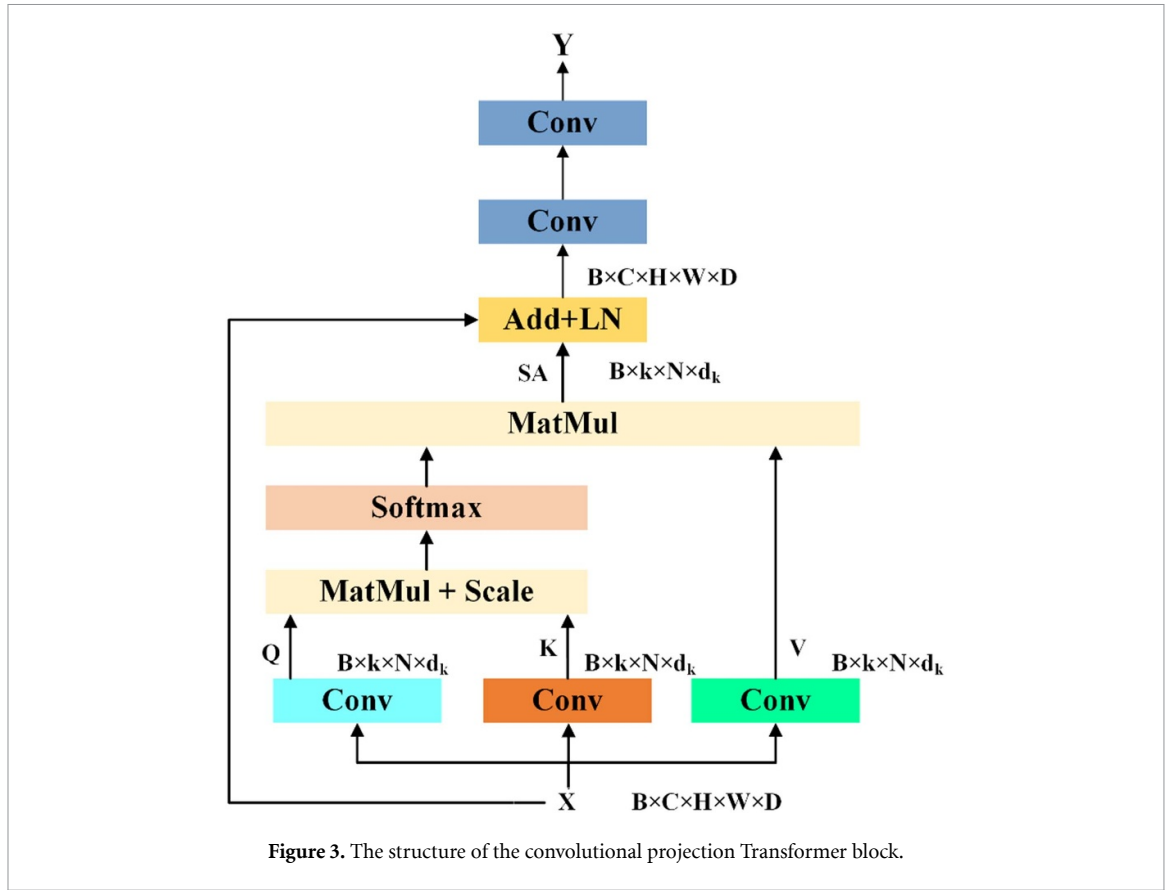
2.1. Architecture of multi-resolution registration network

In this section we describe the particular network architecture used in our method, which is a CNN-Transformer hybrid model integrating the multi-resolution strategy for deformable image registration. Figure 2 depicts the network based on a 3-level Laplacian pyramid framework (Mok and Chung 2020). At the beginning, input image pyramid $\{f_1, f_2, f_3, m_1, m_2, m_3\}$ is created by downsampling with a scale factor $1/2^{3-i}$ ($i = 1, 2, 3$). We then employ a U-shaped network composed of an encoder, a bottleneck module, and a decoder to solve the optimization problem for each pyramid level. For the first pyramid level, the network maps the concatenated input images $[f_1, m_1]$ to deformation field ϕ_1 . For the following two pyramid levels, the deformation field generated from the previous pyramid level ϕ_{i-1} ($i = 2, 3$) is upsampled by a factor of 2 to output $\tilde{\phi}_{i-1}$. Subsequently, the deformed moving image $m_i(\tilde{\phi}_{i-1})$ produced by warping m_i with $\tilde{\phi}_{i-1}$ is concatenated with f_i and $\tilde{\phi}_{i-1}$ to form inputs in current pyramid



level. In particular, feature embeddings at the lower pyramid level are added to the next pyramid level through the skip connection, enabling the expansion of receptive field.

Due to the presence of magnetic field inhomogeneities, we employ a hierarchical cooperative learning paradigm that combines a global main network and a local sub-network. In each pyramid level except the first one, the main network path is responsible for receiving complete images and outputting global deformation fields, while the sub-network path receives divided smaller patches and outputs local deformation fields. As shown in the right side of figure 2, the local sub-network captures feature details of complex cardiac regions by uniformly dividing the complete images, concentrating on the registration



within local small areas. In order to cooperate with main network, outputs of sub-network are sequentially stitched to the size of complete image, resulting in local deformation field. Finally, global deformation field ϕ_i' (in the left side of figure 2) and local deformation field ϕ_i'' are added with $\tilde{\phi}_{i-1}$ to produce the final deformation field ϕ_i for pyramid level i .

2.2. CNN-Transformer based stem

The network components are identical among all the pyramid levels. The encoder is comprised of two $3 \times 3 \times 3$ convolutions with stride 1 and one downsampling layer using $3 \times 3 \times 3$ convolution with stride 2. Symmetrically, the decoder involves one upsampling layer using transpose convolution and two consecutive $3 \times 3 \times 3$ convolutions with stride 1 for predicting deformation field. A skip connection is added from the feature encoder to the feature decoder. Besides, the bottleneck includes one CPTB and five residual blocks with pre-activation structure.

Transformer makes use of self-attention mechanism to capture long-range dependencies. ViT (Dosovitskiy et al 2021) splits an image into a sequence of patches and feeds it into multihead self-attention module for image registration (Xu et al 2023). Swin Transformer limits self-attention computation to non-overlapping local windows by the shifted windows (Liu et al 2021). Although this hierarchical Transformer has linear computational complexity, window-based attention is prone to the lack of global dependencies for complete image. Different from them, we design the Transformer block called CPTB for efficient self-attention computation upon volumetric images. It replaces patch embedding and linear mapping with convolution operation to produce three matrix representations of the Query-Key-Value model, and removes the redundant position embedding. Additionally, instead of multi-layer perceptron, two convolutions with stride 2 are applied to output for modeling local context.

Let X be the input feature map of CPTB with size of $B \times C \times H \times W \times D$, where B is the mini-batch, C is the number of channels, and (H, W, D) is the resolution of input feature. As shown in figure 3, the query matrix Q , the key matrix K , and the value matrix V can be generated by three different convolutions as follow:

$$Q = W^Q \cdot X; K = W^K \cdot X; V = W^V \cdot X \quad (2)$$

Then multi-head operation is performed on N flattened 3D patches with size of $P \times P \times P$, projecting the three matrices into 4D vectors $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{B \times k \times N \times d_k}$, where k is the number of heads, d_k is the number of channels per head, and $C \cdot H \cdot W \cdot D = k \cdot N \cdot d_k$. Therefore, the self-attention is calculated as:

$$SA(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V}. \quad (3)$$

Finally, the SA is added to the input $\mathbf{X}$, and then is processed by one LN (Layer-Norm) and two successive convolution operations. It can be expressed as:

$$\mathbf{Y} = \text{Conv}(\text{Conv}(\text{LN}(\mathbf{SA} + \mathbf{X}))). \quad (4)$$

2.3. Hierarchical cooperative training pattern

Similar to traditional multi-resolution registration, the proposed network can be trained using a coarse-to-fine scheme. To this end, we first train the coarsest pyramid level in an end-to-end manner. The next level at a finer resolution is added to learn image registration problem step by step, where the pre-trained parameters are frozen for M iterations. In particular, for pyramid level $i > 1$, global information from main network properly cooperates with local details from sub-network to improve the performance of deformable registration.

To optimize the network parameters, the overall loss function for training consists of $\mathcal{L}_{\text{sim}}$ and $\mathcal{R}$:

$$\mathcal{L}(f, m, \phi) = \mathcal{L}_{\text{sim}}(f, m \circ \phi) + \lambda \cdot \mathcal{R}(\phi), \quad (5)$$

where λ is a weighting parameter. The first term $\mathcal{L}_{\text{sim}}$ that measures the similarity between the fixed and deformed moving images is local normalized cross-correlation with $\mathcal{N}$ pyramid levels:

$$\mathcal{L}_{\text{sim}}(f, m \circ \phi) = - \sum_{i=1}^{\mathcal{N}} \frac{LNCC_{\omega}(f_i, m_i \circ \phi)}{2^{\mathcal{N}-i}}, \quad (6)$$

where $\omega = 2i + 1$ represents the local window size. The second term $\mathcal{R}$ regularizing the deformation field is defined as:

$$\mathcal{R}(\phi) = \sum_{p \in \Omega} \|\nabla \mathbf{u}(p)\|^2, \quad (7)$$

where $\nabla \mathbf{u}(p)$ denotes the spatial gradient of displacement $\mathbf{u}$ at voxel p , and Ω represents the image domain.

3. Experiments

3.1. Dataset

The proposed method was evaluated on three cardiac image datasets. One hundred short-axis cine MR images from the training set of the MICCAI 2017 Automatic Cardiac Diagnosis Challenge (ACDC17) (Bernard *et al* 2018), were used. This dataset covers five pathology categories: healthy subjects, patients with a systolic heart failure with infarction, patients with dilated cardiomyopathy, patients with hypertrophic cardiomyopathy, and patients with abnormal right ventricle. The cine MR images were acquired on 1.5 T and 3.0 T scanners. The in-plane resolution of the image varies from 0.70 mm to 1.92 mm, and the slice thickness ranges from 5 mm to 10 mm.

The second dataset provided by the MICCAI 2020 the Multi-Centre, Multi-Vendor, Multi-Disease Cardiac Image Segmentation Challenge (M&Ms20) (Campello *et al* 2021), consists of 150 short-axis cine MR images. The subjects were selected among groups of various cardiovascular diseases, including hypertrophic cardiomyopathy, dilated cardiomyopathy, coronary heart disease, abnormal right ventricle, myocarditis and ischemic cardiomyopathy as well as healthy volunteers. The cine MR images were acquired on 1.5 T and 3.0 T scanners from six clinical centers. The in-plane resolution of the image varies from 0.85 mm to 1.45 mm, and the slice thickness ranges from 8 mm to 10 mm.

The third dataset was acquired from the training set of the CMRxMotion Challenge in MICCAI 2022 (CMRxM22) (Wang *et al* 2022). Twenty healthy volunteers were scanned on the same 3.0 T scanner to perform different breath-hold behaviors. They were trained to act in full breath-hold, half breath-hold, free breath, and intensive breath manners. During a single imaging visit, the scan was repeated for four times under above settings. Each subject contributes eight image volumes since the short-axis images at

Table 1. The Dice and Hausdorff distance (HD) of the inter-observer (inter-obs) and intra-observer (intra-obs) variabilities on the ACDC17 dataset. The expert O_1 annotated twice the images (called O_{1a} and O_{1b}) one month apart while the expert O_2 annotated the images once.

Methods	ED				ES			
	LV		MYO		LV		MYO	
	Dice	HD (mm)	Dice	HD (mm)	Dice	HD (mm)	Dice	HD (mm)
O_{1a} vs O_2 (inter-obs)	0.956	5.6	0.870	6.7	0.898	8.1	0.891	7.6
O_2 vs O_{1b} (inter-obs)	0.950	6.2	0.868	7.2	0.895	8.5	0.886	8.0
O_{1a} vs O_{1b} (intra-obs)	0.967	4.0	0.900	5.1	0.941	5.4	0.917	6.0

the end-diastolic (ED) and end-systolic (ES) phases were extracted. The in-plane resolution of the image varies from 0.66 mm to 0.76 mm, and the slice thickness ranges from 9.6 mm to 10 mm.

The ground truth for each image at the ED and ES phases is established through the manual delineation of the left ventricle (LV) and the LV myocardium (MYO) whose contours are called LV Endocardium (LV-Endo) and Epicardium (LV-Epi). For the ACDC17 dataset, these contours were drawn and double-checked by two independent experts (10 and 20 years of experience) who had to reach consensus in case of discordance. Table 1 gives the inter-observer (inter-obs) and intra-obs variabilities (Bernard *et al* 2018) among the manual annotations. In the remaining two datasets, the same standard operating procedures were applied for manual annotations, in order to reduce the inter-obs variability in the contours. For convenience, the data was resampled to a resolution of $1.25 \times 1.25 \times 5$ mm within the ACDC17 dataset, and at a resolution of $1.25 \times 1.25 \times 8$ mm in the M&Ms20 and CMRxM22 dataset. After resampling, the images were cropped to a size of $96 \times 96 \times 16$ from the center of the LV.

3.2. Experimental setup

Deformable image registration was conducted on the CMR images between ED and ES phases without initial global alignment. The image at the ED phase is the fixed image, while the image at the ES phase is the moving image. First, a number of experiments were conducted on the ACDC17 dataset to select the optimal hyperparameters. Subsequently, the performance of the proposed method was evaluated on the M&Ms20 dataset with the tuned hyperparameters. Additionally, deformable image registration was accomplished on the CMRxM22 dataset with the particular imaging condition for further validation. Finally we performed an ablation study to evaluate the contribution of network components.

3.3. Evaluation criteria

To quantitatively assess the quality of deformable image registration, the propagated segmentation of the fixed image can be generated by applying the obtained deformation field to the manual segmentation of the moving image. The Dice Similarity Coefficient (DSC) between the propagated segmentation V_P and the ground truth V_G of the fixed image was calculated as a measure of overlap:

$$\text{DSC}(V_P, V_G) = \frac{2|V_P \cap V_G|}{|V_P| + |V_G|}, \quad (8)$$

where $|\cdot|$ denotes the number of voxels within the segmentation. For surface measure, the Average Symmetric Surface Distance (ASSD) was calculated between the surface of propagated segmentation S_P and the surface of ground truth S_G of the fixed image:

$$\text{ASSD}(S_P, S_G) = \frac{\sum \text{dist}(S_P, S_G) + \sum \text{dist}(S_G, S_P)}{|S_P| + |S_G|}, \quad (9)$$

where $\text{dist}(S_P, S_G)$ is the Euclidean distance of the set of points on S_P to the nearest point on S_G . The Hausdorff Distance (HD) was also calculated between the surface S_P and the surface S_G :

$$\text{HD}(S_P, S_G) = \max\{\max \text{dist}(S_P, S_G), \max \text{dist}(S_G, S_P)\}. \quad (10)$$

To analyze deformation regularity in image registration, the Jacobian determinant of the deformation field $|\mathcal{J}_\phi|$ can be used. As the negative values indicate that the folding and twisting have occurred, we calculated the percentage of voxels with non-positive Jacobian determinant ($|\mathcal{J}_\phi| \leq 0$) (Ashburner 2007). A value of $p < 0.05$ in two-sided Wilcoxon tests (Wilcoxon 1945) indicates a statistically significant difference between two methods.

3.4. Implementation details

The proposed method was implemented using PyTorch (Paszke *et al* 2019) and trained on an NVIDIA TITAN RTX GPU. For the CPTB parameters, the number of channels $C = 16$, the patch size $P = 2$, the number of heads $k = 4$, and the number of channels per head $d_k = 32$ were set. The weighting parameter of loss function λ was set to 0.5. The training process utilized the Adam optimizer with a learning rate of 10^{-4} and a batch size of 1. All registration experiments were run on a computer with an Intel i7-10700 CPU and 64 GB memory.

3.5. Baseline methods

To illustrate the effectiveness of the proposed method, a comparative analysis was conducted, evaluating its performance against seven different image registration methods. This includes four non-learning-based methods and three deep-learning-based methods. The parameter settings of each method are as follows:

- Symmetric normalization (SyN) (Avants *et al* 2008): We used the SyN implementation in the advanced normalization tools package, where local cross-correlation with window size of 7 was employed as the cost function. A step size of 0.25 and Gaussian parameters (3, 0) were set for three resolutions, each with a maximum of 200 iterations.
- NiftyReg (Modat *et al* 2010): We used the NiftyReg package with a multi-threaded CPU version. We employed mutual information as the cost function and bending energy as a regularizer. A regularization weighting of 0.01 and a grid spacing of 8 mm were set for three resolutions, each consisting of 600 iterations.
- Elastix (Shamonin *et al* 2014): We employed a localized version of mutual information as the cost function in software package. A multigrid approach was employed with spacing of 32, 16, and 8 mm between the B-spline control points for the first, second and final resolution, respectively. The number of random samples within 50^3 mm was set to 2000 for three resolutions, with 2000 iterations at resolution. The step size of optimizer was automatically determined using an adaptive stochastic gradient descent strategy (Qiao *et al* 2016).
- PTVReg (Vishnevskiy *et al* 2017): We employed local correlation coefficient as the cost function, is implemented by using a spatial-invariant Gaussian weighting kernel with standard deviation 2.5 mm. The control point grid spacing of free-form deformations was set to $4 \times 4 \times 3$ voxels, and the amount of regularization of 0.1 was used.
- VoxelMorph (Balakrishnan *et al* 2019): The loss function that combined local cross-correlation with window size of 7 and diffusion regularization was employed. A larger architecture called VoxelMorph-2 was selected, and a regularization hyperparameter of $\lambda = 1$ was used. During the training stage, the Adam optimizer with a learning rate of 10^{-4} and a batch size of 1 was utilized for 500 epochs.
- LapIRN (Mok and Chung 2020): A 3-level Laplacian pyramid network framework parameterizing the deformation model with displacement fields was employed. The loss function was local cross-correlation with a smoothness regularization, where window size was set to 3, 5, and 7 for the first, second and final resolution respectively. During the training stage, the hyperparameter $\lambda = 0.5$ and the Adam optimizer with a learning rate of 10^{-4} and a batch size of 1 were set. The default settings were applied to the remaining parameters.
- TransMorph (Chen *et al* 2022): For the Transformer parameters, the patch size was set to 2, the embedding dimension was to be 96, the depths were to be $\{2, 2, 4, 2\}$, the number of heads was set to $\{4, 4, 8, 8\}$, and the shifted window size was set to $\{3, 3, 1\}$. The loss function was a combination of local cross-correlation with window size of 7 and a smoothness regularization. During the training stage, the hyperparameter $\lambda = 1$ and the Adam optimizer with a learning rate of 10^{-4} and a batch size of 1 were set for 500 epochs.

4. Results

4.1. Results on the ACDC17 dataset

In this section, we explored the optimal λ weighting parameter for loss function. The proposed method was then compared to the baseline methods. Experiments including 100 different registration pairs were performed with 5-fold cross validation. Each fold consisted of 70, 10, and 20 pairs (in a 7:1:2 ratio) for training, validation, and testing, respectively.

Figure 4 shows average DSC results of two structures for different weighting parameter $\lambda = \{0.0001, 0.1, 0.5, 1, 2\}$. It is evident that the highest DSC is obtained when $\lambda = 0.5$. Consequently, the weighting parameter of 0.5 was adopted for the proposed method in subsequent experiments.

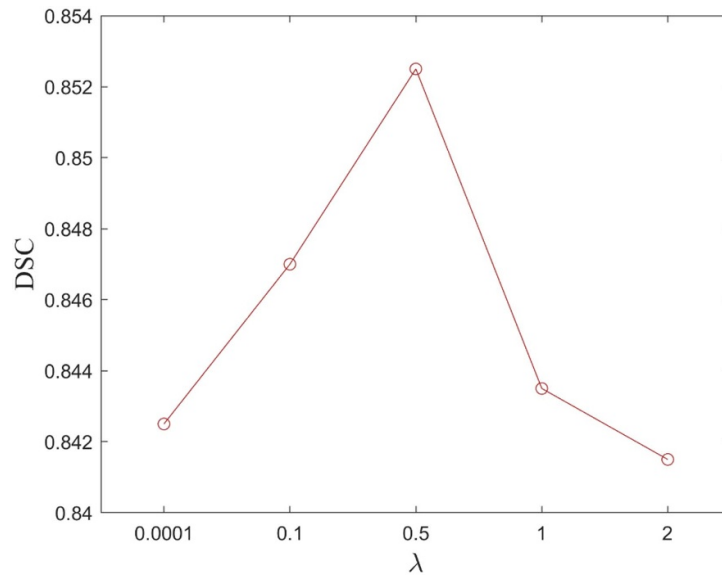


Figure 4. The DSC results of the proposed method with varied hyperparameter λ on the ACDC17 dataset.

Table 2. The mean and standard deviation of quantitative measures using different registration methods on the ACDC17 dataset.

Methods	DSC		ASSD (mm)		HD (mm)		% of $ \mathcal{J}_\phi \leq 0$
	LV	MYO	LV-Endo	LV-Epi	LV-Endo	LV-Epi	
Initial	0.657 ± 0.164	0.518 ± 0.145	4.44 ± 2.28	2.30 ± 1.60	16.3 ± 3.99	12.0 ± 3.16	—
SyN	0.862 ± 0.075	0.741 ± 0.071	1.36 ± 0.75	1.22 ± 0.35	7.83 ± 2.01	7.03 ± 1.66	<0.0001
NiftyReg	0.784 ± 0.160	0.696 ± 0.115	2.87 ± 2.33	1.48 ± 1.34	9.05 ± 1.70	7.40 ± 2.11	0.094 ± 0.198
Elastix	0.862 ± 0.098	0.752 ± 0.068	1.31 ± 0.95	1.15 ± 0.35	7.58 ± 1.90	6.77 ± 1.88	0.277 ± 0.020
PTVReg	0.863 ± 0.094	0.715 ± 0.072	1.81 ± 0.94	1.58 ± 0.89	8.61 ± 2.74	7.96 ± 2.06	0.177 ± 0.286
VoxelMorph	0.871 ± 0.077	0.741 ± 0.084	1.26 ± 0.75	1.16 ± 0.35	7.30 ± 2.11	6.82 ± 1.54	0.176 ± 0.246
LapIRN	0.901 ± 0.059	0.773 ± 0.068	0.96 ± 0.55	0.99 ± 0.31	5.95 ± 1.91	5.80 ± 1.52	0.406 ± 0.191
TransMorph	0.880 ± 0.066	0.747 ± 0.082	1.21 ± 0.71	1.12 ± 0.38	7.12 ± 2.96	6.02 ± 1.85	0.382 ± 0.343
Proposed	0.917 ± 0.043	0.789 ± 0.055	0.79 ± 0.39	0.88 ± 0.21	5.62 ± 1.22	5.51 ± 1.77	0.409 ± 0.153

Table 2 lists the quantitative results of different registration methods. Overall, the proposed method achieved the highest DSC and the lowest surface distance for two structures. LapIRN demonstrates superior performance compared to other baseline methods in terms of DSC, ASSD and HD. Note that SyN produced the smoothest deformation fields.

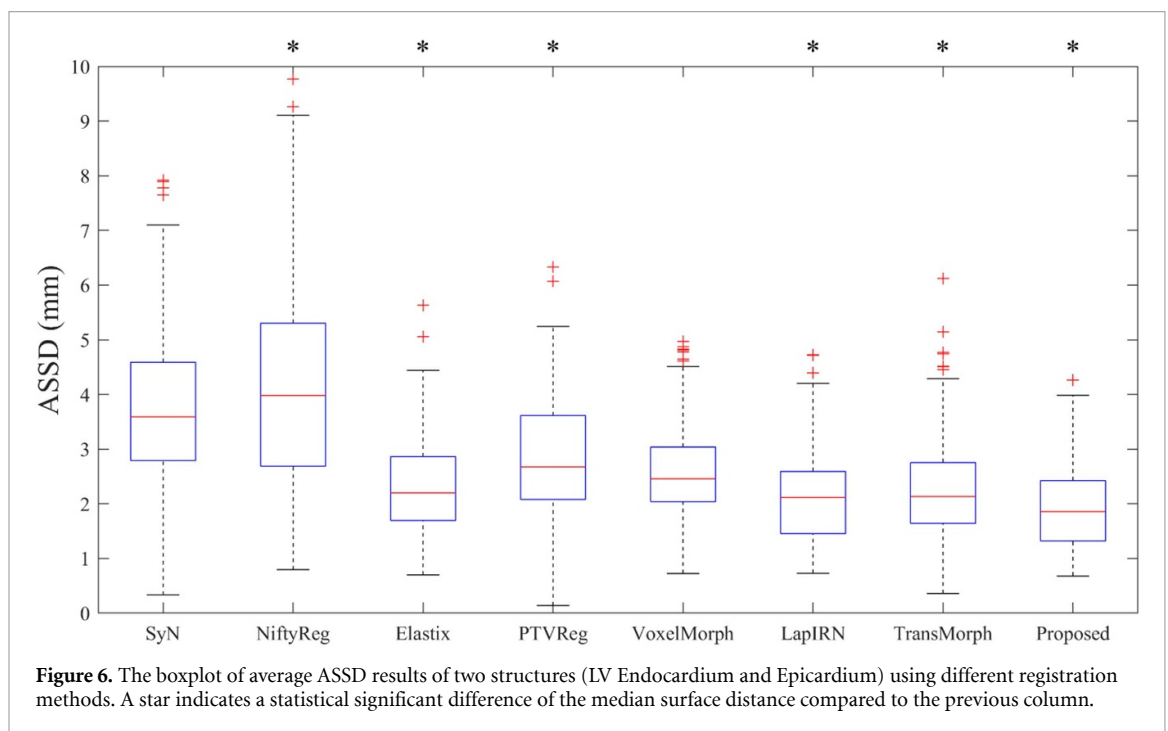
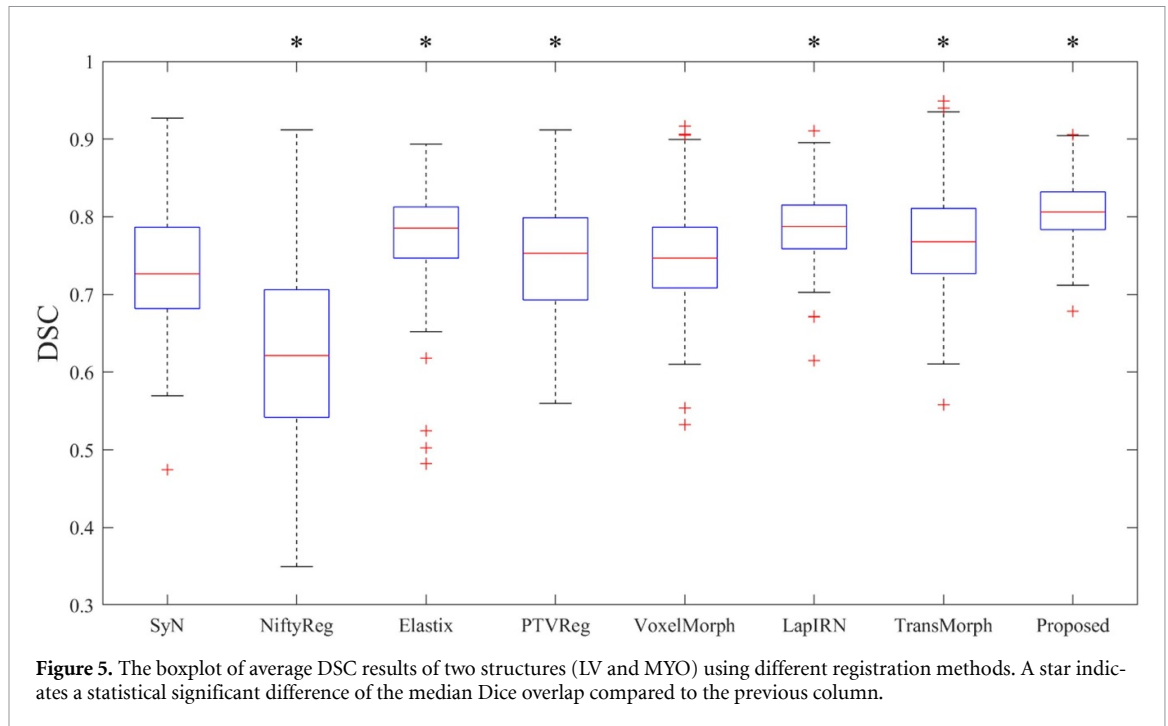
4.2. Results on the M&Ms20 dataset

The proposed method was evaluated on the M&Ms20 dataset using optimized parameters through 5-fold cross validation. Each fold consisted of 105, 15, and 30 pairs (in a 7:1:2 ratio) for training, validation, and testing, respectively. Figures 5 and 6 show average DSC and ASSD results for all methods across two structures. It can be found that the proposed method achieved the highest DSC and the lowest ASSD. The second-best performer in terms of DSC and ASSD is LapIRN. Meanwhile, TransMorph that is a hybrid Transformer-ConvNet model performed better than VoxelMorph.

The quantitative results using different registration methods are summarized in table 3. Compared to LapIRN, the mean Dice overlap of our method increases significantly from 0.863 to 0.884 ($p = 4.35 \times 10^{-6}$) for the LV, and from 0.709 to 0.729 ($p = 1.10 \times 10^{-3}$) for the MYO. The mean ASSD decreases significantly compared to LapIRN from 2.10 mm to 1.88 mm ($p = 3.92 \times 10^{-11}$) for the LV-Endo, and from 2.23 mm to 2.05 mm ($p = 8.09 \times 10^{-8}$) for the LV-Epi. Note that LapIRN obtained the lowest HD for the LV-Epi. Both LapIRN and our method have higher percentage of voxels with non-positive Jacobian determinant than other methods.

4.3. Results on the CMRxM22 dataset

To demonstrate the generalizability of the proposed method, the comparative experiments with the baseline methods were conducted on the CMRxM22 dataset. Due to the availability of data, 67 pairs



of CMR images under different levels of respiratory motion artifacts were used. We randomly selected a trained model from 5-fold validation on the M&Ms20 dataset (section 4.2) for direct predicting the deformation field.

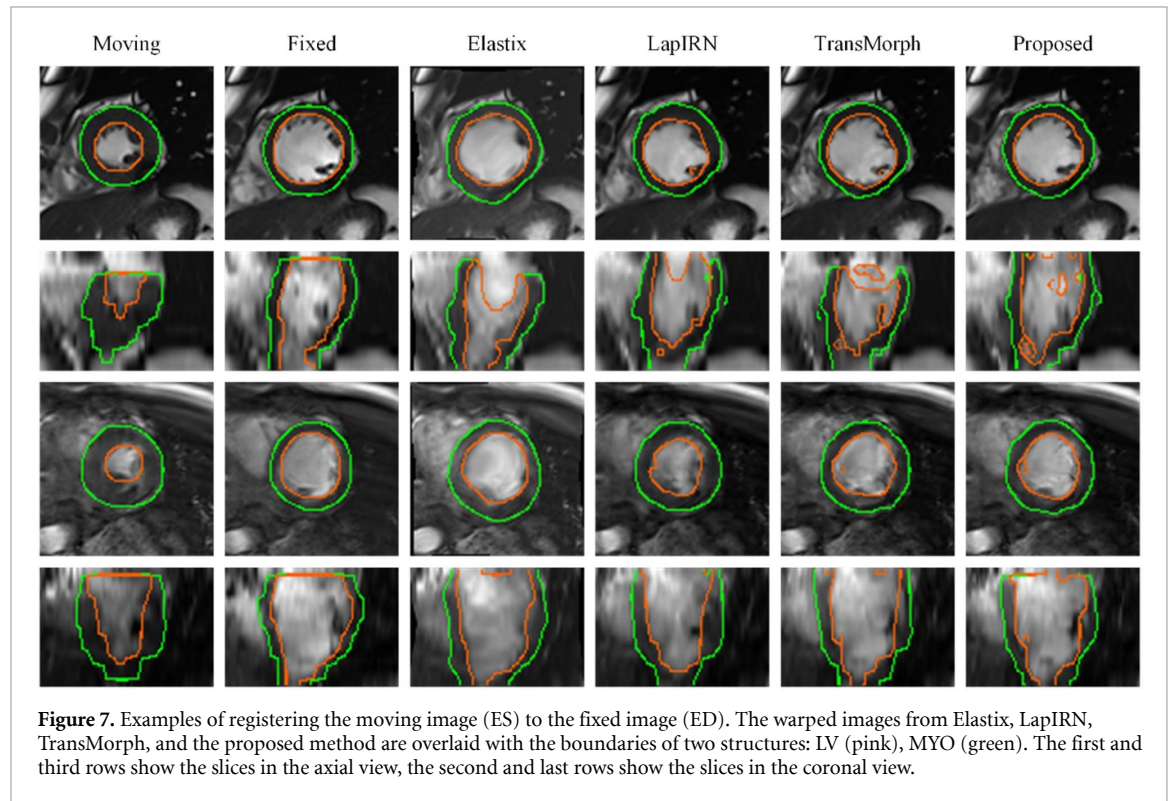
Table 4 lists the quantitative results of different registration methods. Compared with all baseline methods, our method performed the best, with an average DSC of 0.798, ASSD of 1.86 mm and HD of 7.95 mm. Figure 7 illustrates the examples of the warped images produced by Elastix, LapIRN, TransMorph, and our method. It can be observed that our method produces less registration error for the LV in the axial view, and also for the MYO in the coronal view. The displacement field, the circumferential strain and the radial strain of MYO generated from our method are displayed in figure 8.

Table 3. The mean and standard deviation of quantitative measures using different registration methods on the M&Ms20 dataset.

Methods	DSC		ASSD (mm)		HD (mm)		% of $ \mathcal{J}_\phi \leq 0$
	LV	MYO	LV-Endo	LV-Epi	LV-Endo	LV-Epi	
Initial	0.600 $\pm$ 0.126	0.443 $\pm$ 0.142	6.07 $\pm$ 2.10	4.62 $\pm$ 1.78	18.1 $\pm$ 4.28	15.9 $\pm$ 4.29	—
SyN	0.791 $\pm$ 0.081	0.682 $\pm$ 0.088	3.88 $\pm$ 1.57	3.71 $\pm$ 1.63	11.1 $\pm$ 2.35	12.0 $\pm$ 2.27	<0.0001
NiftyReg	0.692 $\pm$ 0.132	0.569 $\pm$ 0.139	4.68 $\pm$ 2.08	3.55 $\pm$ 1.75	13.4 $\pm$ 3.03	11.3 $\pm$ 2.82	0.030 $\pm$ 0.120
Elastix	0.850 $\pm$ 0.092	0.700 $\pm$ 0.055	2.22 $\pm$ 0.83	2.42 $\pm$ 0.91	9.13 $\pm$ 2.09	9.77 $\pm$ 2.34	0.371 $\pm$ 0.446
PTVReg	0.851 $\pm$ 0.089	0.643 $\pm$ 0.087	2.71 $\pm$ 1.13	2.95 $\pm$ 1.34	10.1 $\pm$ 2.21	10.8 $\pm$ 2.49	0.183 $\pm$ 0.497
VoxelMorph	0.820 $\pm$ 0.064	0.675 $\pm$ 0.080	2.67 $\pm$ 0.93	2.50 $\pm$ 0.84	9.89 $\pm$ 2.71	9.68 $\pm$ 2.50	0.020 $\pm$ 0.030
LapIRN	0.863 $\pm$ 0.044	0.709 $\pm$ 0.063	2.10 $\pm$ 0.89	2.23 $\pm$ 0.98	8.73 $\pm$ 2.66	8.05 $\pm$ 2.84	0.498 $\pm$ 0.376
TransMorph	0.854 $\pm$ 0.063	0.684 $\pm$ 0.087	2.21 $\pm$ 1.04	2.35 $\pm$ 1.02	9.08 $\pm$ 2.92	9.71 $\pm$ 2.75	0.014 $\pm$ 0.020
Proposed	0.884 $\pm$ 0.038	0.729 $\pm$ 0.057	1.88 $\pm$ 0.79	2.05 $\pm$ 0.83	7.40 $\pm$ 2.78	8.16 $\pm$ 1.95	0.536 $\pm$ 0.321

Table 4. The mean and standard deviation of quantitative measures using different registration methods on the CMRxM22 dataset.

Methods	DSC		ASSD (mm)		HD (mm)		% of $ \mathcal{J}_\phi \leq 0$
	LV	MYO	LV-Endo	LV-Epi	LV-Endo	LV-Epi	
Initial	0.581 $\pm$ 0.082	0.347 $\pm$ 0.098	6.58 $\pm$ 1.72	4.92 $\pm$ 1.58	17.0 $\pm$ 3.88	16.2 $\pm$ 3.81	—
SyN	0.802 $\pm$ 0.061	0.642 $\pm$ 0.077	3.95 $\pm$ 1.45	3.84 $\pm$ 1.52	12.0 $\pm$ 2.21	11.6 $\pm$ 2.84	<0.0001
NiftyReg	0.673 $\pm$ 0.094	0.482 $\pm$ 0.105	5.25 $\pm$ 1.78	4.10 $\pm$ 1.50	14.2 $\pm$ 2.89	13.3 $\pm$ 2.72	0.051 $\pm$ 0.145
Elastix	0.861 $\pm$ 0.064	0.674 $\pm$ 0.049	2.06 $\pm$ 0.73	2.14 $\pm$ 0.81	8.71 $\pm$ 2.30	8.32 $\pm$ 2.43	0.263 $\pm$ 0.324
PTVReg	0.867 $\pm$ 0.074	0.639 $\pm$ 0.082	2.29 $\pm$ 0.88	2.36 $\pm$ 1.11	8.88 $\pm$ 2.67	9.03 $\pm$ 2.98	0.161 $\pm$ 0.526
VoxelMorph	0.839 $\pm$ 0.048	0.657 $\pm$ 0.063	2.52 $\pm$ 0.78	2.36 $\pm$ 0.69	9.85 $\pm$ 2.33	9.12 $\pm$ 2.45	0.025 $\pm$ 0.036
LapIRN	0.868 $\pm$ 0.033	0.680 $\pm$ 0.052	1.96 $\pm$ 0.74	2.04 $\pm$ 0.83	7.94 $\pm$ 2.60	8.38 $\pm$ 2.56	0.298 $\pm$ 0.184
TransMorph	0.854 $\pm$ 0.051	0.667 $\pm$ 0.067	2.18 $\pm$ 0.93	2.28 $\pm$ 0.85	8.86 $\pm$ 2.76	8.97 $\pm$ 2.55	0.015 $\pm$ 0.016
Proposed	0.892 $\pm$ 0.027	0.703 $\pm$ 0.050	1.78 $\pm$ 0.60	1.94 $\pm$ 0.69	7.82 $\pm$ 2.31	8.07 $\pm$ 2.24	0.303 $\pm$ 0.141



4.4. Results on ablation studies

In this section, ablation experiments were designed on the ACDC17 dataset to verify the effect of various components in our method. Specifically, we begin by examining the effect of CPTB with different numbers of channels on registration performance. The impact of local sub-network on registration performance was also investigated.

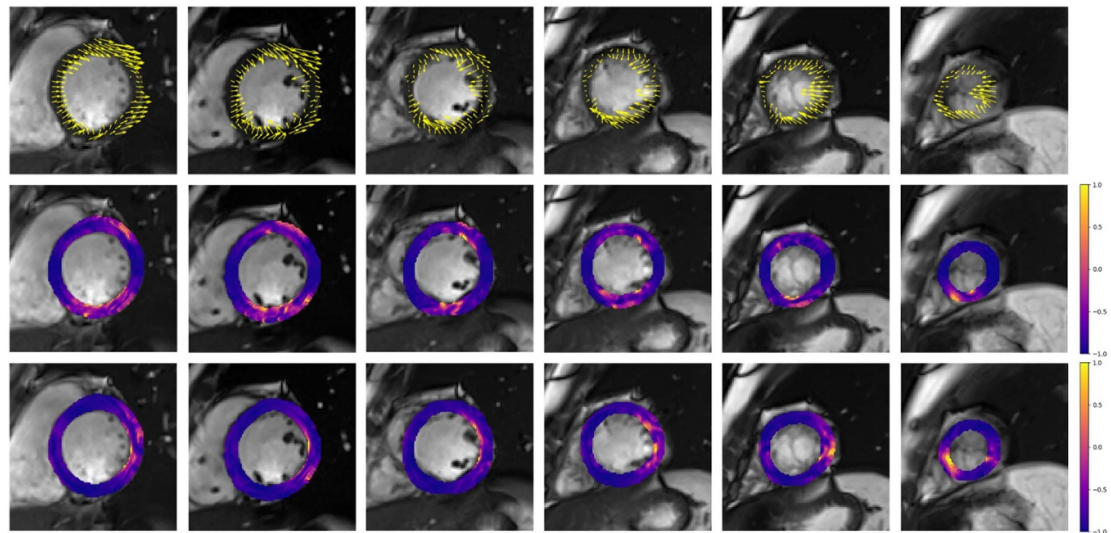


Figure 8. Visualization of motion field (the first row), the circumferential strain (the second row), and the radial strain (the last row) of left ventricle myocardium estimated by the proposed method. The left two columns show basal slices, the middle two columns show mid-ventricular slices, and the right two columns show apical slices.

Table 5. The quantitative measures and model complexities using CPTB with different numbers of channels on the ACDC17 dataset. The DSC, ASSD and HD are computed by average values of two structures (LV and MYO).

Number of channels	DSC	ASSD (mm)	HD (mm)	Params (M)
$C = 8$	0.852	0.84	6.01	0.15
$C = 16$	0.853	0.84	5.57	0.58
$C = 32$	0.851	0.84	5.80	2.30

Table 5 lists the quantitative results and model complexities obtained from the ablation studies using CPTB with different numbers of channels $C = \{8, 16, 32\}$. Compared to $C = 8$, the average DSC of $C = 16$ increases slightly from 0.852 to 0.853, and the average HD of $C = 16$ decreases from 6.01 mm to 5.57 mm. However, the model with $C = 32$ has the lowest DSC and maximum number of trainable parameters. As a result, we choose $C = 16$ for the proposed method.

The purpose of local sub-network is to mitigate the issue of magnetic field inhomogeneities. We explore the impact of local sub-network on deformable registration by removing the CPTB component. Compared to LapIRN, our method without the CPTB component resulted in a mean DSC improvement of 0.01, and a mean ASSD reduction of 0.08 mm. The results of the ablation experiments provide evidence for the effectiveness of local sub-network.

5. Discussion and conclusion

In this paper, we have introduced a multi-resolution CNN-Transformer registration framework for cardiac motion estimation. For large and complex deformations, the convolution operation is utilized in place of patch embedding and linear mapping of ViT, which has the ability to establish long-range spatial correspondences. To deal with magnetic field inhomogeneities, a cooperative learning paradigm is employed for information fusion of global main network and local sub-network. Finally, deformable registration of CMR images can be optimized in a coarse-to-fine fashion to facilitate accurate estimation of motion field.

As can be seen from table 1, the Dice score oscillates between 0.86 and 0.96, and the HD between 4 mm and 8.5 mm. In general, the bSSFP sequence can obtain good accuracy thanks to the high quality of images, which also contributes to reducing the inconsistency between manual delineation. Regarding the HD values, the main contributing factor is the large inter-slice thickness. In contrast, the proposed method yielded the average Dice of 0.917 for the LV and 0.789 for the myocardium. The average HD was 5.62 mm for the endocardium and 5.51 mm for the epicardium. These results fall within the inter-obs variabilities for most metrics (except for the Dice of the myocardium), indicating a promising performance.

From experimental results on three datasets, it is clear that the proposed framework with a comparable number of parameters achieves a higher accuracy than the state-of-the-art methods. Owing to high computational complexity of ViT, the operations on CPTB implementation would reduce network parameters, keeping registration performance (see table 5). The average inference time per image pair is 0.092s. The Dice scores of the myocardium, which seem to fall below the inter-obs scores, were not as good as those of the LV. To further assess the performance of the proposed method relative to the experts variability, approaches such as atlas-based segmentation and validation on more extensive datasets can be adopted. The application of our approach to other registration tasks, such as registration between CT and MR scans (Blendowski *et al* 2021), is an area we intend to pursue in the future. A limitation of our method is that a large percentage of voxels with non-positive Jacobian determinant is produced. One possible option for the smoother deformation field is the incorporation of conditional instance normalization (Mok and Chung 2021).

For use in the clinic, further improvements might be possible. One viable option is to integrate anatomical priors into the loss function for constraints of the deformation field. These priors in form of keypoint correspondences (Hering *et al* 2021) and anatomical labels (Balakrishnan *et al* 2019) are only required for training. Another issue of involving prior information is groupwise registration using deep learning framework for cardiac motion estimation (Fechter and Baltas 2020). Compared to pairwise registration, groupwise registration performs the optimization of network parameters for all time phases of cine sequence of CMR simultaneously. The cyclic motion known as *a priori* is easy to be added into the loss function for periodic deformation. In future research, we plan to investigate the performance of groupwise registration based on the proposed framework.

In conclusion, based on the evaluation and comparison with several standard approaches, our proposed method has demonstrated superior performance for the challenging problem of cardiac motion estimation.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary information files).

Data and code availability

Data is openly available. You can get them at the website given in the references. The code is freely available at: <https://github.com/xslu-scuec/DIR-MRVIT>.

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